

Exercise 5

1. Use Gaussian elimination with partial pivoting to solve the following linear system.

We can get that by dividing matrices:

```
clc;
clear all;
close all;
A = [1,1,-1;1,2,4;2,-1,2];
b = [1,2,3]';
X=A\b
```

```
X = 3x1
    1.2632
   -0.0526
    0.2105
```

2. Solve the following linear system by using Chasing method

$$\begin{cases} x_1 + x_2 = 5 \\ 2x_1 - x_2 + 5x_3 = -9 \\ 3x_2 - 4x_3 + 2x_4 = 19 \\ 2x_3 + 6x_4 = 2 \end{cases} \quad (1)$$

```
%Chasing method
clc;
clear all;
close all;
```

```

A=[1,2,0,0;2,3,-1,0;0,4,2,-1;0,0,2,-4];
b=[7,9,10,12];
b=b';
ab=[A,b];%a为系数矩阵, b为右端项
[R,C]=size(ab);
for i=1:R-1
    for j=i+1:R

        ab(j,i:C)=ab(j,i:C)-ab(i,i:C)*ab(j,i)/ab(i,i);%每一行消去首行元素
    end
end
AB=ab;
for i=1:R
    ab(i,:)=ab(i,:)/ab(i,i);
end
A
b
fprintf("高斯消去法后的增广矩阵: ")
AB
fprintf("追赶法后的增广矩阵: ")
ab
X=zeros(R,1);%解空间
X(R,1)=ab(R,C);
for i=R-1:-1:1
    X(i,1)=ab(i,C)-X(i+1,1)*ab(i,i+1);
end
fprintf("解为:")
X

```

A = 4×4

1	2	0	0
2	3	-1	0
0	4	2	-1
0	0	2	-4

```
b = 4×1
    7
    9
   10
   12
```

高斯消去法后的增广矩阵:

```
AB = 4×5
    1    2    0    0    7
    0   -1   -1    0   -5
    0    0   -2   -1  -10
    0    0    0   -5    2
```

追赶法后的增广矩阵:

```
ab = 4×5
    1.0000    2.0000    0    0    7.0000
         0    1.0000    1.0000    0    5.0000
         0         0    1.0000    0.5000    5.0000
         0         0         0    1.0000   -0.4000
```

解为:

```
X = 4×1
    7.4000
   -0.2000
    5.2000
   -0.4000
```

3. Find the $\|X\|_1$, $\|X\|_2$, $\|X\|_\infty$ norms the following vectors

(1) $X = (3, -4, 0, \frac{3}{2})^T$,

```
clc;
clear all;
close all;
X = [3,-4,0,3/2]';
Y = [2,1,-3,4]';
```

```

[a,b,c] = Get_normal_form(X);
fprintf("求得的第一范式为：")
a
fprintf("求得的第二范式为：")
b
fprintf("求得的无限范式为：")
c

function [abs_1,abs_2,abs_inf] = Get_normal_form(x)
    abs_1 = 0;
    abs_2 = 0;
    for i=1:length(x)
        abs_1 = abs_1 + abs(x(i));
        abs_2 = abs_2 + x(i)^2;
    end
    abs_2 = sqrt(abs_2);
    abs_inf = max(abs(x));

end

```

```

X
求得的第一范式为：
a = 8.5000
求得的第二范式为：
b = 5.2202
求得的无限范式为：
c = 4

```

(2) $Y = (2, 1, -, 4)^T$

